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FORM TP 21134

CANDIDATE - PLEASE NOTE!

You must sign below and return this booklet with the Answer Sheet. Failure to do so may result in disqualification.

Signature _____

TEST CODE **002401**

MAY/JUNE 2001

CARIBBEAN EXAMINATIONS COUNCIL

**SECONDARY EDUCATION CERTIFICATE
EXAMINATION**

PHYSICS

Paper 01 – General Proficiency

1 $\frac{1}{4}$ hours

11 JUNE 2001 (a.m.)

READ THE FOLLOWING DIRECTIONS CAREFULLY

1. In addition to this test booklet, you should have an answer sheet.
2. Each item in this test has four suggested answers lettered (A), (B), (C), (D). Read each item you are about to answer and decide which choice is best.
3. On your answer sheet, find the number which corresponds to your item and shade the space having the same letter as the answer you have chosen. Look at the sample item below.

Sample Item

The SI unit of length is the

- (A) newton
- (B) metre
- (C) kilogram
- (D) second

Sample Answer

(A) ● (C) (D)

The best answer to this item is "metre" so answer space-(B) has been shaded.

4. If you want to change your answer, be sure to erase your old answer completely and fill in your new choice.
5. When you are told to begin, turn the page and work as quickly and as carefully as you can. If you cannot answer an item, omit it and go on to the next one. You can come back to the harder item later. Your score will be the total number of correct answers.
6. Figures are not necessarily drawn to scale.
7. You may do any rough work in this booklet.
8. The use of non-programmable calculators is allowed.
9. This test consists of 60 items. You will have 75 minutes to answer them.
10. Do not be concerned that the answer sheet provides spaces for more answers than there are items in this test.

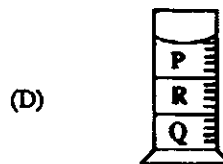
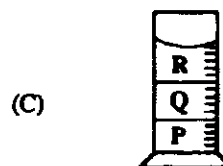
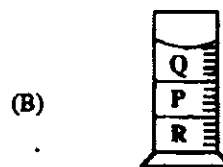
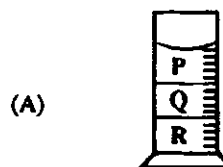
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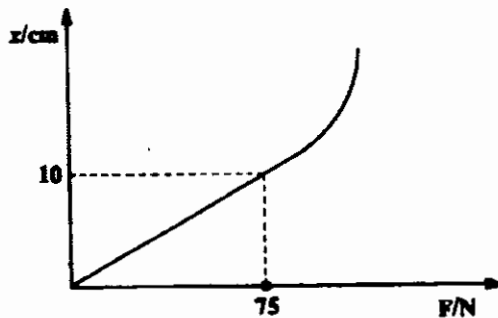
1. Which of the following is NOT a fundamental quantity?
- (A) Temperature
(B) Heat energy
(C) Length
(D) Time
2. Which of the following would be correct?
- I. 1 milliamperes = 10^{-6} A
II. 1 megohm = $10^6 \Omega$
III. 1 microcoulomb = 10^{-3} C
- (A) I only
(B) II only
(C) I and II only
(D) I, II and III
3. The instrument MOST suitable for measuring the external diameter of a test tube is the
- (A) micrometer screw gauge
(B) vernier calipers
(C) metre rule
(D) set square
4. A student finds the time for a ball to fall 50.0 cm in oil is 14.8 s. She divides the distance by the time to find the speed, and the calculator displays 3.3783783. The best way to write this value is
- (A) 3.4 cm s^{-1}
(B) 3.38 cm s^{-1}
(C) 3.378 cm s^{-1}
(D) $3.3783783 \text{ cm s}^{-1}$
5. Which of the following means of measurement would produce the LEAST ACCURATE result assuming the instrument used is working properly?
- (A) Using a metre rule to measure the length of a desk
(B) Using vernier calipers to find the thickness of a piece of paper
(C) Using a lever arm balance to find the mass of a book
(D) Using a stop watch to time a 400m race
6. Three immiscible liquids P, Q and R, have densities which differ. Q is denser than P but less dense than R. Which of the following diagrams shows how the liquids settle in the measuring cylinder?



7. The force which keeps the moon in orbit is

- (A) magnetic
- (B) centrifugal
- (C) reactive
- (D) gravitational

8.



The diagram above shows a simple extension (x)/force (F) graph for a light spring. Which of the following statements would be true?

- I. The elastic limit of the spring was exceeded.
- II. The spring obeyed Hooke's law over its entire extension.
- III. The force per unit extension in the elastic region was 7.5 N cm^{-1} .

- (A) I only
- (B) I and III only
- (C) II and III only
- (D) I, II and III

9. An object is NOT accelerating if

- (A) its direction of motion is changing but its speed is not
- (B) its velocity is changing
- (C) it is stationary for a moment in mid-air
- (D) it has no unbalanced force acting on it

10. Which of the following could be used to find the average speed of an object?

- (A) $\frac{\text{Change in velocity}}{\text{Time taken}}$
- (B) $\frac{\text{Change in displacement}}{\text{Time taken}}$
- (C) $\frac{\text{Distance travelled}}{\text{Time taken}}$
- (D) Distance travelled $\times$ Time taken

11. A ball attached to a string is whirled round at a constant speed in a horizontal circle. It is true to say that

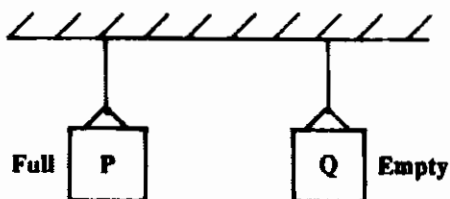
- (A) the velocity is constant
- (B) there are no forces acting on the body
- (C) the motion is accelerated
- (D) the ball will drop vertically downward if the string breaks

12. A book of weight 12 N is resting on a table. The force that the table exerts on the book is

- (A) 0 N
- (B) greater than 0 N but less than 12 N
- (C) 12 N
- (D) greater than 12 N

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13.



Two identical covered tins are suspended by strings from a beam. Tin P is full of sand while Tin Q is empty. You can tell which is which by giving each a push. The full tin will require a larger push to start swinging. This is because Tin P is

- (A) greater weight than Tin Q
- (B) greater mass than Tin Q
- (C) greater momentum than Tin Q
- (D) greater volume than Tin Q

14. The kinetic energy of a body of mass, m , and velocity, v , is

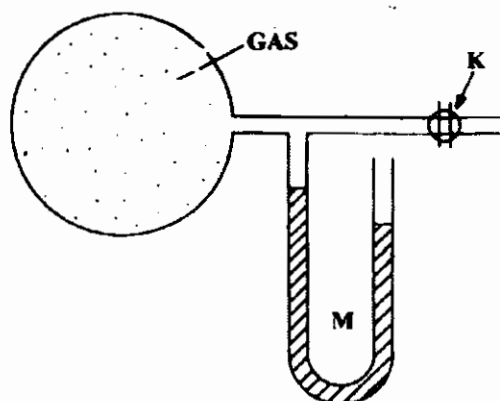
- (A) $\frac{m}{v}$
- (B) mv
- (C) $\frac{m^2}{v}$
- (D) $\frac{1}{2}mv^2$

15. The rate of which of the following determines the power output of a machine?

- I. Doing work
- II. Converting energy
- III. Changing temperature

- (A) I only
- (B) III only
- (C) I and II only
- (D) I, II and III

16. The globe shown in the diagram below contains a gas the pressure of which is measured with a manometer, M.

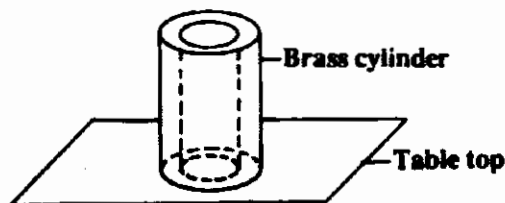


Which of the following statements would be correct?

- I. The pressure of the gas is less than that of the atmosphere.
- II. If the right hand side of the manometer tube were closed the reading would be more accurate.
- III. In order to bring the two menisci in the manometer to the same level one could open the stop-cock, K.

- (A) I only
- (B) II only
- (C) I and II only
- (D) I and III only

17.

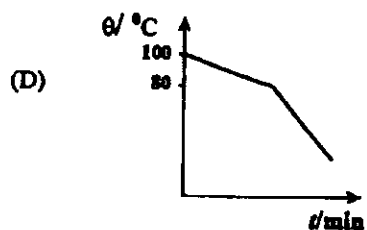
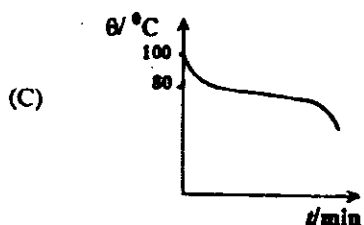
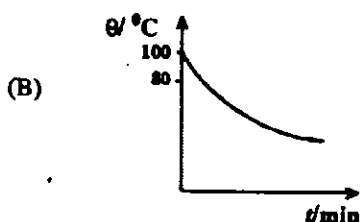
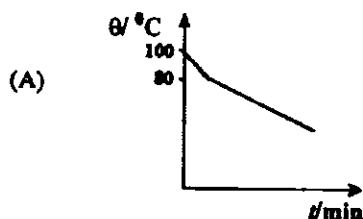


A brass cylinder is standing on a table as shown in the diagram above. It is heated electrically until its temperature reaches 150°C . Which of the following properties increases as its temperature rises?

- (A) Density
- (B) Volume
- (C) Pressure it exerts on the table
- (D) Specific heat capacity

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18. Some molten naphthalene at 100°C is allowed to cool down at room temperature. If naphthalene has a melting point of 80°C , which of the following BEST represents the cooling curve?



19. The bombardment by the molecules of a gas on the walls of its containing vessel determines its

(A) mass
(B) volume
(C) pressure
(D) temperature

20. The unit for specific latent heat of fusion is

(A) J
(B) J kg^{-1}
(C) J K^{-1}
(D) $\text{J kg}^{-1} \text{K}^{-1}$

- 21.



The figure above shows a piece of ice wrapped in copper gauze and submerged in a glass test tube of water. The water when heated at the top boils long before the ice is melted. The MAIN reason for this is that

- (A) the test tube is a poor conductor of heat
(B) copper is a good conductor of heat
(C) ice is less dense than water
(D) water is a poor conductor of heat

22. Which of the following would cause the hot air above a fire to rise?

- I. Increase in the speed of the air molecules
II. Expansion of the air
III. Decrease in the density of the air

- (A) I only
(B) III only
(C) II and III only
(D) I, II and III

23. Which of the following occurs when atoms arrange themselves in a regularly repeated pattern?

- (A) Crystallisation
(B) Evaporation
(C) Diffusion
(D) Condensation

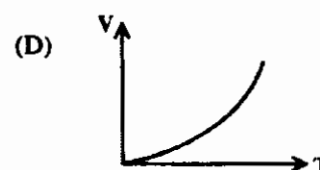
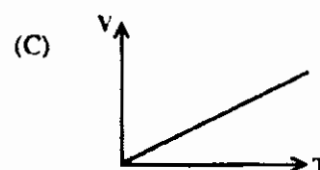
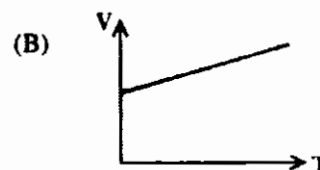
24. Which of the following is true of evaporation?

- (A) It occurs throughout the liquid at no definite temperature.
- (B) It occurs at the surface of the liquid at no definite temperature.
- (C) It occurs at the surface of the liquid at a definite temperature.
- (D) It occurs throughout the liquid at a definite temperature.

25. The 'glass-house' effect refers to the glass-house acting as a heat trap. This is a direct result of

- (A) short wavelength infra-red rays which are unable to penetrate glass
- (B) long wavelength infra-red rays radiated from the sun which are unable to penetrate glass
- (C) short wavelength infra-red rays radiated from the objects within the glass-house which are unable to penetrate glass
- (D) long wavelength infra-red rays radiated from the objects within the glass-house which are unable to penetrate glass

26. Which of the following graphs is obtained if the volume of a fixed mass of gas, at a constant pressure, is plotted against its Kelvin temperature?



27.



A longitudinal wave is shown in the above diagram. Which of the following represents a wavelength?

- (A) PQ
- (B) QS
- (C) QR
- (D) QT

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28. Which of the following statements about sound is NOT correct?

- (A) Sound is transmitted as transverse waves.
- (B) Sound may be produced by vibrating systems.
- (C) Sound does not travel through a vacuum.
- (D) Sound travels more slowly than light.

Item 29 refers to the following.

Two parallel scratches are made on a slide which is otherwise painted black. The slide is mounted so that the scratches are parallel to the straight filament of a red lamp.

29. An experiment of this type was first performed by

- (A) Newton
- (B) Young
- (C) Huygens
- (D) Foucault

30. Which of the following is/are involved in a production of fringes on a Young's double-slit experiment?

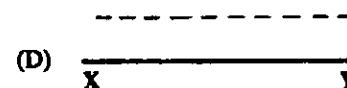
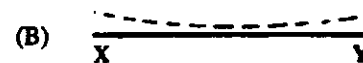
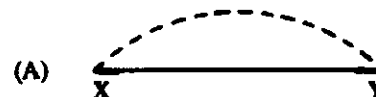
- I. Diffraction
- II. Refraction
- III. Interference

- (A) I only
- (B) III only
- (C) I and III only
- (D) I, II and III only

31.



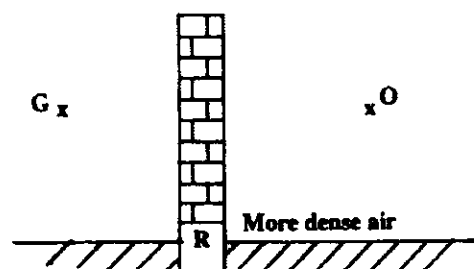
The figure above shows a wave-front about to reach a reflecting surface XY. Which of the following would represent the wavefront after reflection?



32. Sound waves are classified as longitudinal since they

- (A) cause the particles of the medium to move along the line of propagation of the waves
- (B) require a material medium for transmission
- (C) may be produced by vibrating bodies
- (D) cause the particles of the medium to move at right angles to the line of propagation of the waves

33.



In the diagram above, R is a tall narrow wall. The report of a gun, fired at G, is heard by an observer at O. Which of the following could account for this fact?

- I. Diffraction takes place round the obstacle.
- II. The sound waves are longitudinal.
- III. Refraction takes place towards the ground.

- (A) I only
- (B) I and III only
- (C) II and III only
- (D) I, II and III

34. Which of the following would be involved in a production of fringes in the well-known double-slit experiment?

- I. Reflection
- II. Refraction
- III. Interference

- (A) I only
- (B) III only
- (C) I and III only
- (D) I, II and III only

35. Which of the following demonstrates the straight line propagation of light?

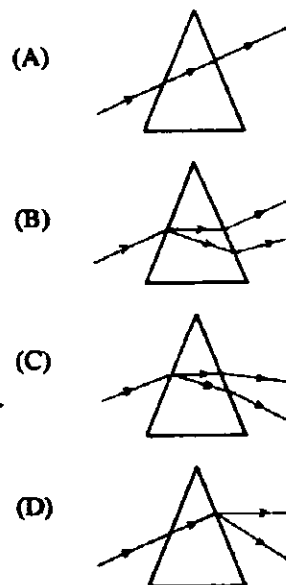
- (A) The spreading of light when it passes through a narrow opening
- (B) The dispersion of light by a prism
- (C) A parallel beam of light coming to a point focus after reflection from a spherical mirror
- (D) The formation of shadows

36. For which of the following does the phenomenon of refraction offer an explanation?

- I. The apparent bending of a stick in water
- II. Swimming pools looking shallower than they really are
- III. Formation of a mirage

- (A) I and II only
- (B) I and III only
- (C) II and III only
- (D) I, II and III

37. Which of the diagrams BEST represents the passage of a beam of white light through a triangular glass prism?



38. A converging lens produces an image which is the same size as the object. It is true to say that the

- (A) object is at the principal focus
- (B) object is at infinity
- (C) image is virtual
- (D) image is inverted

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39. Which of the following would be true of the image formed by the lens of a camera?

- I. It is virtual.
- II. It is real.
- III. It is diminished.
- IV. It is upright

- (A) I only
- (B) II and III only
- (C) I and IV only
- (D) IV only

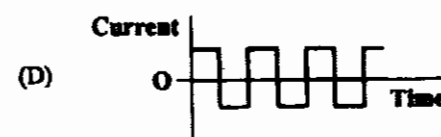
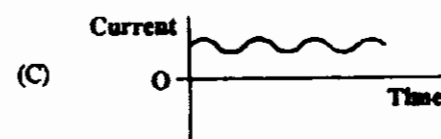
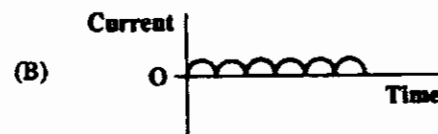
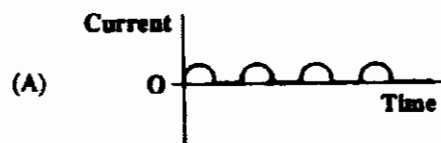
40. Which of the following is NOT one of the ways in which the strength of the magnetic field near a solenoid (long coil) carrying a current can be increased?

- (A) Increasing the resistance of the coil
- (B) Increasing the current in the coil
- (C) Increasing the number of turns per unit length of the coil
- (D) Placing a soft iron core inside the coil

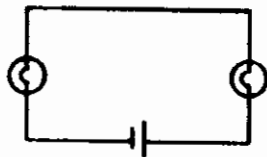
41. Which of the following formulae (in which the symbols have their usual meaning) is a mathematical expression of Ohm's Law?

- (A) $V = IR$
- (B) $P = VI$
- (C) $Q = It$
- (D) $E = Pt$

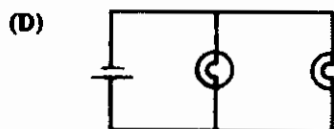
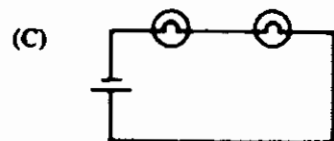
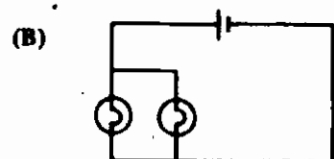
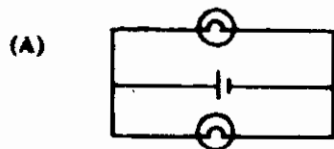
42. Which of the following graphs illustrates an ALTERNATING current?



- 10 -



43. The diagram above shows an electrical circuit with a cell and two filament bulbs. Which of the following circuits is electrically the same as the circuit above?



44. Which of the following statements concerning ammeters is correct?

	Its resistance	Circuit connection
(A)	Low	in series
(B)	Low	in parallel
(C)	High	in parallel
(D)	High	in series

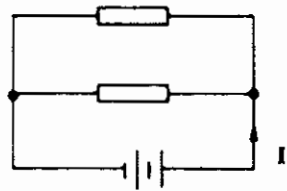
45. Which of the following is true of a secondary cell?

- (A) It is formed by connecting two or more primary cells.
(B) It can be recharged by passing a direct current in the same direction as it delivers current.
(C) It can be recharged by passing a direct current in the opposite direction to which it delivers current.
(D) It can be recharged by 'topping up' with dilute sulphuric acid.

46. In an electric water heater the metal parts, not including the heater circuit, are connected to earth by a conductor. The reason for this is to

- (A) prevent a short circuit
(B) set up an electric field
(C) set up a potential difference between the metal parts and earth
(D) ensure that the metal parts are kept at the same potential as earth

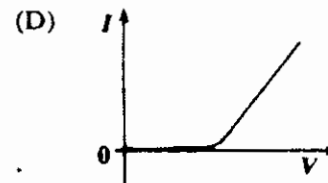
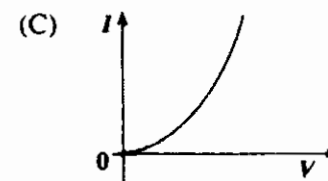
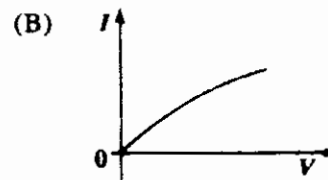
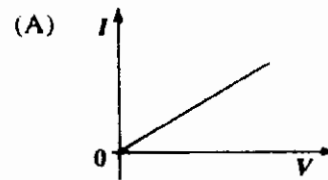
47.



In the parallel circuit shown above, the total current supplied by the battery

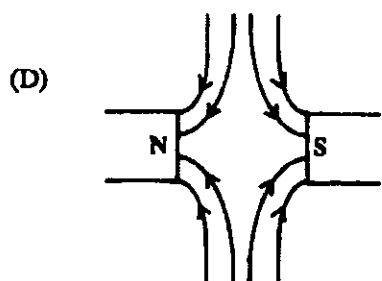
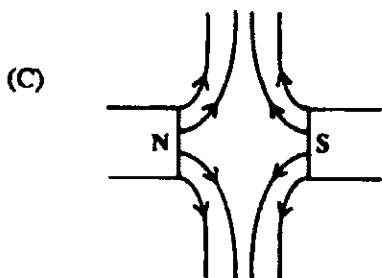
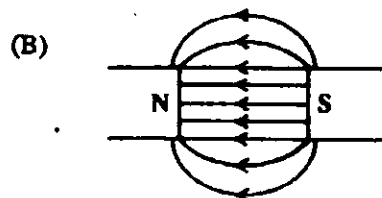
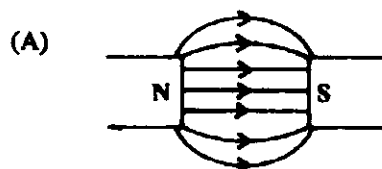
- (A) goes through each of the resistors
- (B) goes through the largest resistor only
- (C) goes through the smallest resistor only
- (D) equals the sum of the currents in the two resistors

Item 48 refers to the following I - V graphs for various conducting media under certain conditions.



48. Which graph illustrates the relationship between current and applied p.d. for a lamp filament?

49. A north pole and a south pole of two equally strong magnets are brought near each other. If the earth's magnetic field is ignored, the magnetic field lines between the poles would look like

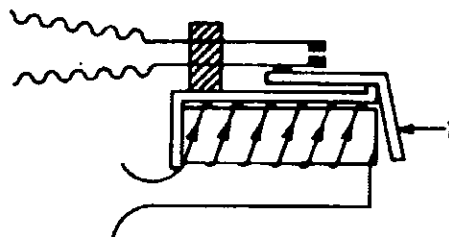


50. A semi-conductor diode produces half-wave rectification of alternating current. Which of the following statements would be true?

- I. The current obtained has a constant value.
- II. The current obtained flows only in one direction.
- III. There are periods when no current flows from the source.

- (A) I only
(B) II only
(C) I and III only
(D) II and III only

51.



The diagram above shows a typical relay. The part, labelled P, must be made of

- (A) iron
(B) copper
(C) plastic
(D) brass

52. Which of the following is true when a magnet is moved relative to a coil?

- I. The induced current is in the same direction as the change causing it.
- II. The greater the number of turns of coil the smaller the induced e.m.f.
- III. The faster the magnet moves relative to the coil the greater the induced e.m.f.
- IV. The stronger the magnet the greater the induced e.m.f.

- (A) I and IV only
(B) I and III only
(C) II and III only
(D) III and IV only

53. A conductor, rotating in a uniform magnetic field, induces maximum instantaneous current when the conductor cuts the magnetic field lines at
- (A) 30°
(B) 45°
(C) 90°
(D) 180°
54. An ideal transformer has 6 000 primary turns and 4 000 secondary turns. The secondary current is
- (A) $1/2$ of the primary current
(B) $2/3$ of the primary current
(C) $3/2$ times the primary current
(D) twice the primary current
55. α -particles pass through a piece of gold foil. It is observed that some of the α -particles are deflected from their original direction, a very small number of them being deflected through large angles. This experiment provides evidence that
- (A) the atom of gold is very small
(B) the atom of gold has a small positively charged nucleus
(C) an α -particle is repelled by the electrons in the gold atom
(D) α -particles are much larger than atoms
56. According to the Rutherford-Bohr model of a neutral, stable atom, if n = number of neutrons, p = number of protons and e = number of electrons in the atom, then for all elements
- (A) $n = e$
(B) $p = e$
(C) $n + e = p$
(D) $n + p = e$
57. Which of the following statements may be used to define the term 'half-life'?
- I. It is half the time required by a nucleus to decay.
II. It is the time for half of the nuclei of a sample to decay.
III. Time taken for sample undergoing a random decay process, to decay to half its original value
- (A) I only
(B) I and II only
(C) II and III only
(D) I, II and III
58. ${}^x_{52}\text{Te} \rightarrow {}^0_{-1}\text{e} + {}^{137}_{53}\text{I}$
- The equation above represents the radioactive decay of a certain nucleus. The values of x and y are.
- | | x | y |
|-----|-----|-----|
| (A) | 137 | 53 |
| (B) | 137 | 52 |
| (C) | 137 | 51 |
| (D) | 136 | 53 |
59. From which of the following can a sheet of lead of thickness 1 mm provide good protection?
- I. α -particles
II. β -particles
III. γ -rays
- (A) I and II only
(B) I and III only
(C) II and III only
(D) I, II and III
60. Which of the following persons is associated with nuclear energy?
- (A) Thompson
(B) Bohr
(C) Chadwick
(D) Einstein

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.